Andrew Grady Johnson, Ph.D.

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Summary

Computational scientist studying fine-grained human mobility and urban systems using machine learning, spatial analysis, and agent-based modeling. Intelligence Community Postdoctoral Fellow at the University of Pennsylvania, collaborating with Oak Ridge National Laboratory on computational models of human movement using large-scale mobility data. Broader research background spans ecological and evolutionary theory, experimental microbiology, and computational analysis of policy text. Additional technical work includes developing local LLM applications, including a two-stage retrieval-augmented generation system integrated into LM Studio, and deploying large open-weight models on memory-constrained hardware.

Education

Aug. 2017 – Sep. 2024 **University of Pennsylvania**, *Ph.D., M.Sc.* in Biology
Dissertation title: *Community Ecology of the Gut Microbiome.*
Advisor: Dustin Brisson

Aug. 2008 – Dec. 2012 **North Carolina State University**, *B.S.* in Animal Science

Professional Experience

Sep. 2024 – Current **Penn Institute for Urban Research**
Position: Intelligence Community Postdoctoral Fellow

- Build and evaluate computational models of fine-grained human mobility in cities using large-scale mobility data, street-segment geometry, built-environment features, and temporally stratified prediction tasks.
- Develop machine learning workflows for tabular spatial prediction, including feature engineering, train-test evaluation, stacked ensembles, model comparison, and interpretation of temporally varying predictors.
- Use agent-based modeling in Python to simulate pedestrian decision-making and test how local movement assumptions scale into aggregate spatial patterns.
- Established an active research collaboration between Penn IUR and Oak Ridge National Laboratory focused on human mobility modeling and urban-scale movement prediction.

Aug. 2023 – May 2024 **Wharton, University of Pennsylvania**
Position: Research Associate

- Applied stacked ensemble machine learning models to predict zoning restrictiveness from built-environment features, reducing prediction error by more than 8%.
- Developed an LLM-based pipeline to extract structured variables from unstructured municipal zoning ordinances and policy documents.
- Communicated technical methods and results to interdisciplinary and non-technical executive-level stakeholders to support decision-making.

Aug. 2022 – May 2023 **Perry World House, University of Pennsylvania**
Position: Graduate Fellow

- Worked at the intersection of scientific and policy challenges in modern biodefense and biosecurity, writing policy-facing essays on pathogens of enhanced pandemic potential and dual-use research of concern (PEPP/DURC).

Professional Experience (continued)

Jul. 2022 – Sep. 2024 **Adaptive Cities Working Group**
Position: Scientific Advisor (Ecology and Simulation Design)
- Interdisciplinary working group, in partnership with the Rockefeller Foundation and the Brookings Institution, focused on using digital twins to address the intersection of cities, heat waves, and health.
- Supported technical project design and implementation planning for applied urban modeling workflows.

Aug. 2017 – Sep. 2024 **Department of Biology, University of Pennsylvania**
Position: Doctoral Researcher
- Designed, built, and validated a controlled experimental microbiome system for studying host-associated microbial community assembly, stability, and perturbation response.
- Conducted experimental microbiology work involving microbial culture, system maintenance, contamination control, sampling, DNA extraction, sequencing-oriented workflows, and quantitative analysis.
- Applied ecological and evolutionary theory to questions of coexistence, community assembly, invasion, and stability in host-associated microbial systems.

Research Publications

1. A. Johnson, J. Hira, and D. Brisson, "The gut microbiome can influence female mating behavior in *D. melanogaster*," *manuscript in preparation*, 2026.
2. A. Johnson, J. Pfrommer, and D. Brisson, "Phylogenetic determinants of coexistence in the gut microbiome of *D. melanogaster*," *manuscript in preparation*, 2026.
3. A. Johnson, K. Sparks, N. Johnson, G. Thakur, and T. Smyth, "Built-environment predictors of street-segment pedestrian activity in new york city," in *International Conference on Social Computing, Behavioral-Cultural Modeling and Prediction and Behavior Representation in Modeling and Simulation*, Springer, 2026, pp. 209–218.
4. N. Johnson and A. Johnson, "Network exposure to solved homicides reduces future violence," *manuscript in preparation*, 2026.
5. A. Renninger, A. Johnson, D. Stanek, D. Voith, and S. Wachter, "Predicting zoning restrictions with remote sensing and machine learning," *manuscript in preparation*, 2026.
6. A. Johnson, "Community ecology of the gut microbiome," Ph.D. dissertation, University of Pennsylvania, 2024.
7. B. Raynor et al., "Movement patterns of free-roaming dogs on heterogeneous urban landscapes: Implications for rabies control," *Preventive Veterinary Medicine*, vol. 178, p. 104 978, 2020.
8. R. Castillo-Neyra et al., "Reemergence of canine rabies in complex urban environments: Lessons from an outbreak in Arequipa, Peru," in *American Journal of Tropical Medicine and Hygiene*, vol. 97, 2017, pp. 578–578.

Scientific/Technical Leadership

- Led cross-disciplinary research teams of up to six people, coordinating research design, analysis plans, troubleshooting, trainee work, and manuscript development.
- Mentored 14 trainees across master's, post-baccalaureate, undergraduate, and high-school levels, including experimental design, data analysis, writing, and career development.

Skills

Programming	Python, R, SQL, Java, LaTeX, HTML
Technology	Git/GitHub, Jupyter, Anaconda, LM Studio, Strata, Overleaf, NetLogo
LLM Systems & Applications	Retrieval-augmented generation (RAG), semantic vector search, embedding models, reranking, LM Studio plugin integration, LLM API integration

Skills (continued)

LLM Architecture & Inference	Local LLM deployment, hardware-constrained inference, model quantization, sparse mixture-of-experts (MoE) architectures, memory-mapped embeddings
Machine Learning & Modeling	Random Forest, LightGBM, XGBoost, stacking, ensembling, agent-based modeling, regression, hypothesis testing, probability theory, spatial forecasting / spatial modeling, temporal dependence modeling
Data & Research Workflows	Web scraping, corpus construction and cleaning, LLM-based information extraction from unstructured text, feature engineering, model evaluation, data preprocessing, API integration, regex/text parsing
Communication	Technical writing and presentations for academic, executive, and public audiences

Fellowships, Awards, & Achievements

2024 **Intelligence Community Postdoctoral Research Fellowship**, Enhanced Computational Modeling of Human Navigation in Urbanized Environments (Reference: ICPD-2024-05), $300,000

Schmidt Science Fellowship Nominee, University of Pennsylvania

2022 **Perry World House Graduate Fellowship**, University of Pennsylvania

SASGov Research Student Grant, University of Pennsylvania, $2,000

2021 **GAPSA-Provost Fellowship for Interdisciplinary Innovation**, Penn Provost's Office, $6,000

2020 **President**, Penn Biology Graduate Group

Teece Dissertation Research Fellowship, University of Pennsylvania, $2,500

SASGov Research Student Grant, University of Pennsylvania, $2,000

Selected Technical Projects

Local LLM Workflows & Retrieval Augmented Generation (Python)

- Built a two-stage RAG system using a web-scraped and cleaned literary corpus, combining semantic vector search with secondary reranking to retrieve contextually relevant examples.
- Integrated the system into LM Studio as a plugin, enabling corpus-grounded manuscript analysis directly within the chat interface.
- Configured and ran a 75.8 GB Qwen3.8-Flash-Next model on consumer hardware (64 GB RAM, 12 GB VRAM) using Strata, taking advantage of its sparse MoE architecture and SSD-backed n-gram embeddings to overcome memory constraints.

Executive Action Summary Mailer (Python)

- Converts executive actions (executive orders, memorandums, etc.) from the White House webpage to text files, passes them to OpenAI's API for a structured summary, and emails it.

NYT Letter Boxed Solver (Python)

- Searches for a minimal chain of words that use every letter provided, where each word starts with the last letter of the previous word and no adjacent letters come from the same side. Built a custom word bank by parsing text files from Project Gutenberg.

Personal Hobbies

Jan. 2025 – Current **Writing**, Flash cnf and fiction, poetry. Writing published in River Teeth.

Aug. 2019 – Sep. 2024 **Judo**, Penn Judo Club/Main Line Judo Club, Member